Expanded GS-PnP SSDU tuning

Dedicated preparation and experiment report

Current status: Completed validation sweep, frozen parameters and retrospective 450-image test.

1 Executive summary

The previous ground-truth-free GS-PnP selection was $\lambda = 0.1$, denoiser $\sigma = 0.075$ and initial $\tau_0 = 100/3$. Each was at the top of the previous finite search range. The five-image held-out measurement MSE was 0.0043654339. We evaluated 27 preregistered candidate configurations, including the old winning control and 26 previously untested combinations. Selecting among them must use only archived validation measurements, never image ground truth or the previously inspected 450-image test scores.

2 Tuning algorithm

Use the same five NPL-noisy validation images: CIGS 0733, Organic 0008, Perovskite 0283, Reference 0768 and Silicone 0589. For each image, retain the first 4,096 coarse rows of the 13,107 acquired Hadamard measurements in every reconstruction fold. Divide the remaining 9,011 acquired fine rows into the four previously frozen disjoint SSDU holdouts.

For candidate $\theta = (\lambda, \sigma, \tau_0)$, reconstruct from fit rows Θ_{ik} and score held-out rows Λ_{ik}:

$$L(\theta) = \frac{1}{5}\sum_{i=1}^{5}\frac{1}{4}\sum_{k=1}^{4}\frac{\|A_{\Lambda_{ik}}\widehat{x}_{\theta,ik} - y_{\Lambda_{ik}}\|_2^2}{|\Lambda_{ik}|}.$$

Keep the original normalised PCM operator and summed half-squared residual fidelity, without a fold-size rescaling. Use the same pretrained GS-DRUNet and paper-final z image.

Frozen parameter grid

Parameter	Candidate values
λ	0.10, 0.20, 0.40
Denoiser σ	0.075, 0.10, 0.15
Initial τ_0	33.3333, 50, 75

Freeze 100 maximum accepted steps, 20 line-search attempts, $\gamma = 0.1$, $\eta = 0.9$ and stopping tolerance 10^{-6}. Tune λ and τ_0 independently. Record accepted iterations, rejections, final τ, effective $\lambda\tau$, and time per reconstruction.

3 Step-by-step plan

1. Verify the frozen acquisition hashes, original four fold masks, solver equivalence and data-fidelity scale. Freeze source hashes and the candidate grid.
2. Reuse the original 20 fold scores of `gs_018` under control ID `gs_exp_001`, keeping original SHA256 provenance.
3. Score 26 new configurations across five images and four folds: 520 new GS reconstructions. Each can take up to 100 accepted iterations, with additional backtracking work.
4. Record all held-out losses and solver diagnostics. Exclude configurations with any failed fold. If more than nine candidates fail, withhold selection.
5. Select the minimum balanced SSDU loss, breaking ties within 10^{-10} by grid order. Freeze the candidate and every result hash.

6. Check whether the winning parameters are still at the expanded upper bounds. Only after freezing, consider a separately labelled retrospective image-quality comparison.
7. Refresh this report with actual scores and numerical diagnostics. Do not tune to test PSNR.

4 Experiment running

The expanded sweep completed on 9 November 2026. The computing GPU was tuni's RTX 4090, using an isolated CUDA UUID and deterministic cuBLAS/cuDNN. The original float64 noisy measurements and mask indices are reused, while neural solvers receive their required float32 values. The scoring code does not load targets. Case-level JSON files allow safe resumption. The old tuned-5 and tuned-50 selection records and the 450-image test benchmark are preserved.

5 Results and deliverables

Run status: 24 valid, 3 invalid, 0 untested/incomplete candidates. The candidate search is complete and the selection is frozen. The validation-selected winner is `gs_exp_002`, with $\lambda = 0.100$, $\sigma = 0.075$, $\tau_0 = 50.0000$, and held-out MSE 0.00436100.

Candidate	λ	σ	τ_0	SSDU MSE	Status
`gs_exp_001`	0.10	0.075	33.3333	0.00436543	reused control
`gs_exp_002`	0.10	0.075	50.0000	0.00436100	complete
`gs_exp_003`	0.10	0.075	75.0000	0.00436156	complete
`gs_exp_004`	0.10	0.100	33.3333	0.00437060	complete
`gs_exp_005`	0.10	0.100	50.0000	0.00437426	complete
`gs_exp_006`	0.10	0.100	75.0000	0.00437598	complete
`gs_exp_007`	0.10	0.150	33.3333	0.00441525	complete
`gs_exp_008`	0.10	0.150	50.0000	0.00441581	complete
`gs_exp_009`	0.10	0.150	75.0000	0.00441612	complete
`gs_exp_010`	0.20	0.075	33.3333	0.00437101	complete
`gs_exp_011`	0.20	0.075	50.0000	0.00437115	complete
`gs_exp_012`	0.20	0.075	75.0000	0.00436703	complete
`gs_exp_013`	0.20	0.100	33.3333	0.00439591	complete
`gs_exp_014`	0.20	0.100	50.0000	0.00439712	complete
`gs_exp_015`	0.20	0.100	75.0000	0.00439825	complete
`gs_exp_016`	0.20	0.150	33.3333	0.00444078	complete
`gs_exp_017`	0.20	0.150	50.0000	0.00444064	complete
`gs_exp_018`	0.20	0.150	75.0000	0.00444091	complete
`gs_exp_019`	0.40	0.075	33.3333	0.00438429	complete
`gs_exp_020`	0.40	0.075	50.0000	0.00438489	complete
`gs_exp_021`	0.40	0.075	75.0000	–	invalid
`gs_exp_022`	0.40	0.100	33.3333	0.00441816	complete
`gs_exp_023`	0.40	0.100	50.0000	0.00441543	complete
`gs_exp_024`	0.40	0.100	75.0000	–	invalid
`gs_exp_025`	0.40	0.150	33.3333	0.00447255	complete
`gs_exp_026`	0.40	0.150	50.0000	0.00447335	complete
`gs_exp_027`	0.40	0.150	75.0000	–	invalid

The dedicated folder holds the original grid, preflight, resumable numerical code, case-level results, frozen selection and an editable report in both TeX and PDF.

Expanded sweep: measured validation outcome

The 27 candidates were predeclared before the GPU evaluation. The original control risk was 0.004365434, while the minimum validation-only held-out risk was 0.004361004 for `gs_exp_002`. This is a 0.10 percent reduction in the SSDU loss relative to the original control. It is a difference in *measurement prediction error*, not necessarily image reconstruction quality.

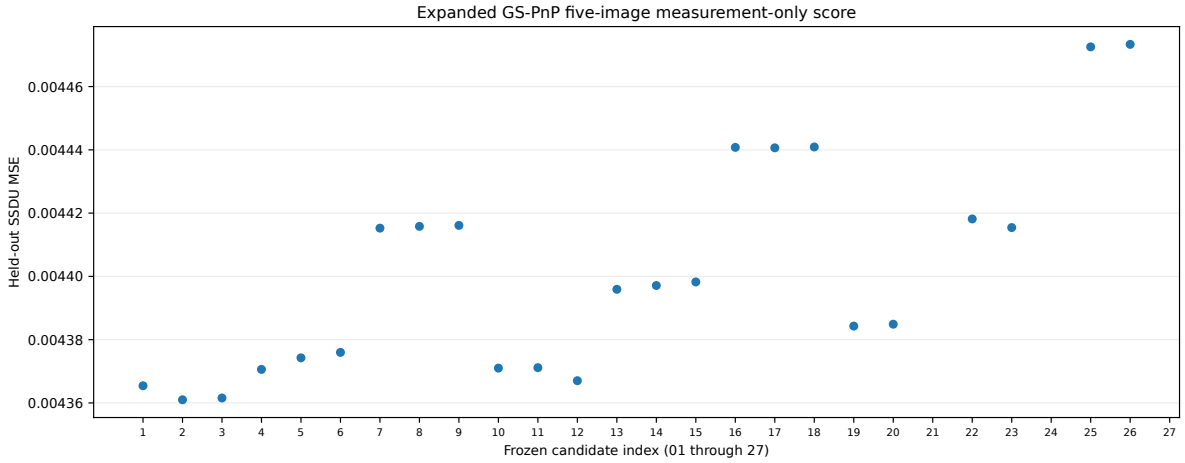


Figure 1: Validation-only SSDU loss across completed candidates.

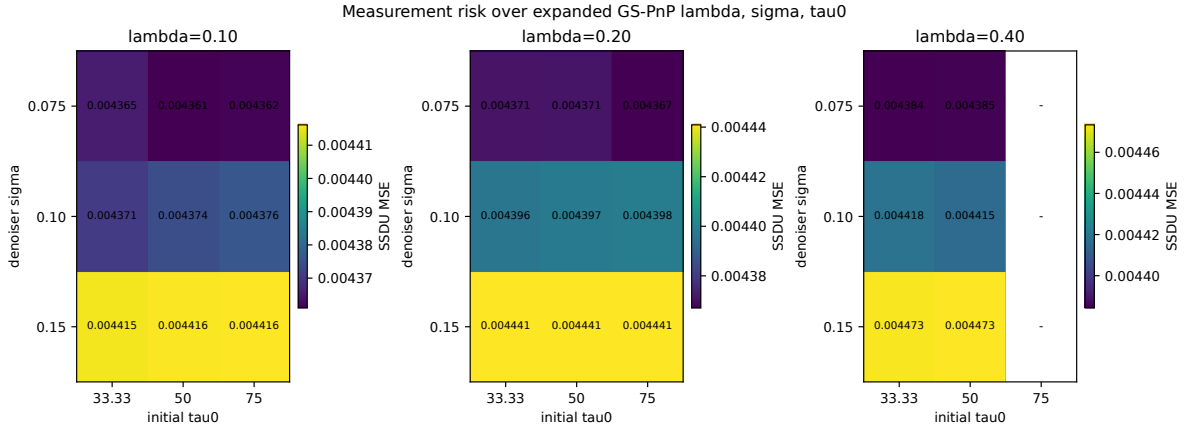


Figure 2: Expanded lambda, sigma and initial step-size slices of the validation-only risk.

The frozen winning parameters are

$$\lambda = 0.100, \qquad \sigma = 0.075, \qquad \tau_0 = 50.0000.$$

The selection used only the original five validation images and their fixed holdout masks. No test-image ground truth or previous test metrics influenced the selection. There were 3 numerically invalid configurations out of 27. The winner's upper-bound flags are: lambda: no, sigma: no, tau0: no.

Category	Original SSDU MSE	Expanded SSDU MSE	Change
CIGS	0.00392860	0.00390345	-0.00002515
Organic	0.00427257	0.00427257	-0.00000000
Perovskite	0.00469730	0.00469710	-0.00000020
Reference	0.00406831	0.00406816	-0.00000015
Silicone	0.00486039	0.00486374	+0.00000335

Table 2: Per-category validation measurement prediction errors: one image and four held-out folds per category.

Invalid candidate	Validation case	Failure reason
gs_exp_021	Perovskite_0283	No GS iteration accepted
gs_exp_021	Silicone_0589	No GS iteration accepted
gs_exp_024	Perovskite_0283	No GS iteration accepted
gs_exp_027	Perovskite_0283	No GS iteration accepted

Table 3: Numerically invalid higher-value candidates; they were excluded rather than silently assigned a favourable SSDU loss.

Candidate	λ	σ	τ_0	**SSDU risk**
gs_exp_002	0.100	0.075	50.000	0.00436100
gs_exp_003	0.100	0.075	75.000	0.00436156
gs_exp_001	0.100	0.075	33.333	0.00436543
gs_exp_012	0.200	0.075	75.000	0.00436703
gs_exp_004	0.100	0.100	33.333	0.00437060
gs_exp_010	0.200	0.075	33.333	0.00437101
gs_exp_011	0.200	0.075	50.000	0.00437115
gs_exp_005	0.100	0.100	50.000	0.00437426
gs_exp_006	0.100	0.100	75.000	0.00437598
gs_exp_019	0.400	0.075	33.333	0.00438429

Table 4: Top ten valid candidates ranked by frozen five-image SSDU loss.

Config	**Folds**	**Accepted iters**	**Rejections**	$\boldsymbol{\lambda\tau}$	**s/solver**
gs_exp_002	20	48.0	11.8	1.466	1.054
gs_exp_001	20	47.9	8.1	1.450	0.986

Table 5: Selected versus original GS control. Means over 20 validation folds. A faster method may require fewer accepted iterations or less work per iteration/backtracking.

Retrospective 450-image PCM evaluation

After freezing the expanded SSDU winner, we applied it to the exact same stored fixed 20 percent NPL-noisy acquisitions used for the previous 450-image comparisons. For each test image, the sampled Hadamard-mask and noisy measurement SHA256 hashes were verified against the archived baseline. We did not select or modify hyperparameters after examining reconstruction metrics.

The expanded configuration gives mean PSNR 32.563 dB, compared with 32.590 dB for the original SSDU-5 GS-PnP and 35.110 dB for the supervised-validation oracle. The paired expanded-minus-original PSNR difference is -0.027 dB (sample SD 0.531 dB), while the difference against the oracle is -2.547 dB. All test parameters were frozen before scoring these images.

The expanded configuration gave higher PSNR on 349 of 450 matched images, with a median paired change of +0.127 dB. However, 43 images lost over one dB, producing the negative mean paired change despite most cases improving. An eight-case hardware parity audit on the RTX 4090, including large gain and loss cases, reproduced the earlier RTX 4070 original configuration PSNR to within 0.00009 dB. Thus the hardware difference does not explain those selected outliers; full 450-image hardware parity has not been tested.

Method	PSNR, dB	SSIM	HaarPSI	MSE
Zero-filled	24.97 ± 1.97	0.4823 ± 0.0527	0.5513 ± 0.0520	3.53e-03 ± 1.62e-03
SPGL1-BP	25.42 ± 1.53	0.4353 ± 0.0354	0.6452 ± 0.0772	3.06e-03 ± 1.18e-03
SPGL1-BPDN	29.37 ± 3.41	0.7161 ± 0.1036	0.6581 ± 0.0994	1.55e-03 ± 1.18e-03
PnP-ADMM oracle	35.75 ± 3.43	0.8970 ± 0.0770	0.8814 ± 0.0619	3.54e-04 ± 2.43e-04
GS-PnP oracle	35.11 ± 3.64	0.8709 ± 0.0871	0.8628 ± 0.0735	4.18e-04 ± 2.87e-04
TV	30.03 ± 1.94	0.7212 ± 0.0553	0.7844 ± 0.0504	1.10e-03 ± 5.47e-04
UTV-C60	32.30 ± 2.93	0.8450 ± 0.0882	0.7699 ± 0.1101	7.28e-04 ± 4.53e-04
PnP-ADMM tuned-5	35.75 ± 3.43	0.8970 ± 0.0770	0.8814 ± 0.0619	3.54e-04 ± 2.43e-04
GS-PnP tuned-5	32.59 ± 2.70	0.7570 ± 0.0967	0.8208 ± 0.0775	6.63e-04 ± 3.91e-04
GS-PnP expanded-5	32.56 ± 2.65	0.7558 ± 0.0948	0.8201 ± 0.0771	6.63e-04 ± 3.84e-04

Table 6: Mean ± sample SD over 450 paired images, calculated on raw (unclipped) reconstructions. Ten method labels are shown, including the original SSDU-5 and newly expanded-5 GS-PnP.

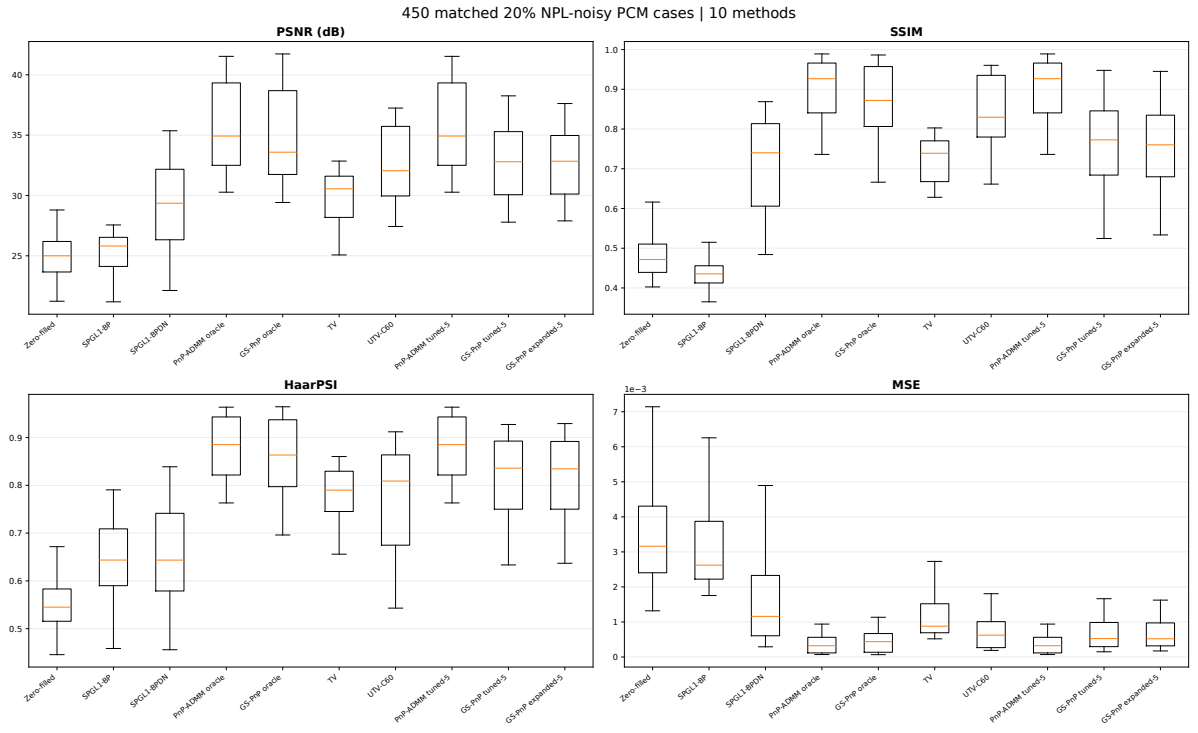


Figure 3: Paired 450-image score distributions for ten methods. Outliers are not drawn in the boxplot, but all 450 values contribute to the descriptive statistics.

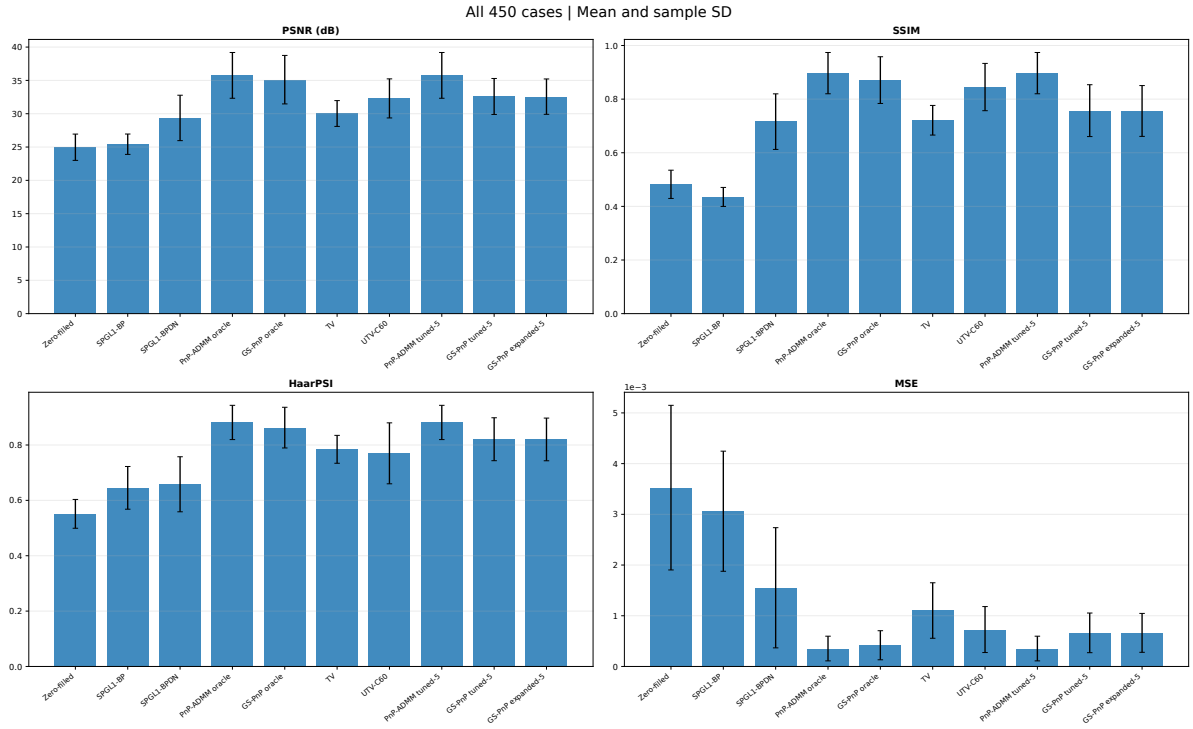


Figure 4: Mean and standard deviation for all ten methods across 450 images.

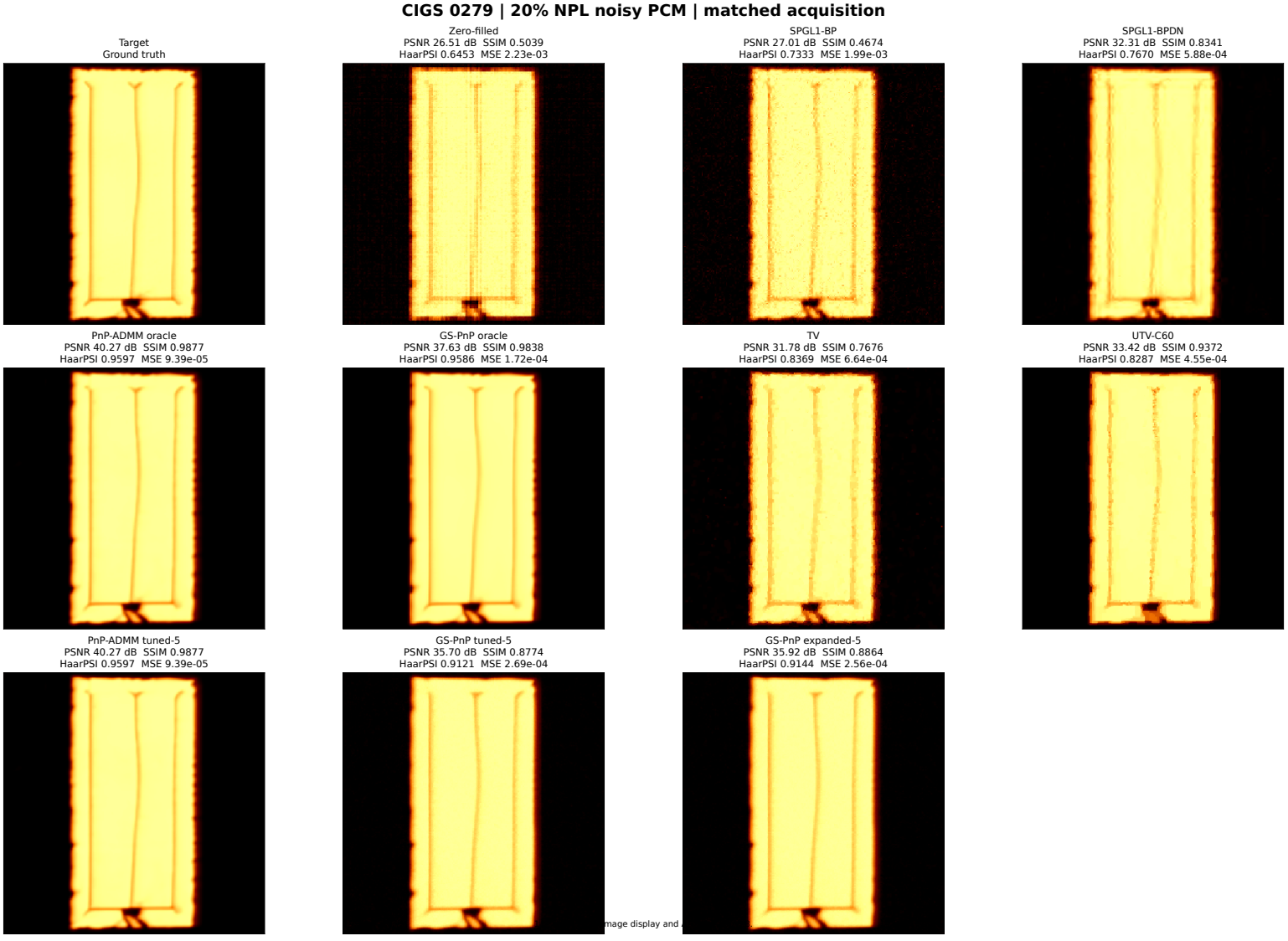
Results by device category

Category	ZF	SPGL1	BPDN	ADMM	GS oracle	TV	C60	GS SSDU-5	GS expanded
CIGS	25.71	26.49	31.10	39.30	37.89	31.16	33.04	35.31	35.25
Organic	25.49	26.37	32.05	38.90	38.94	31.50	35.72	34.96	34.83
Perovskite	27.06	26.43	31.60	35.48	35.35	31.33	32.61	31.70	31.72
Reference	24.25	24.39	26.72	32.14	31.72	28.30	29.63	29.73	29.73
Silicone	22.34	23.44	25.38	32.95	31.65	27.87	30.49	31.25	31.27

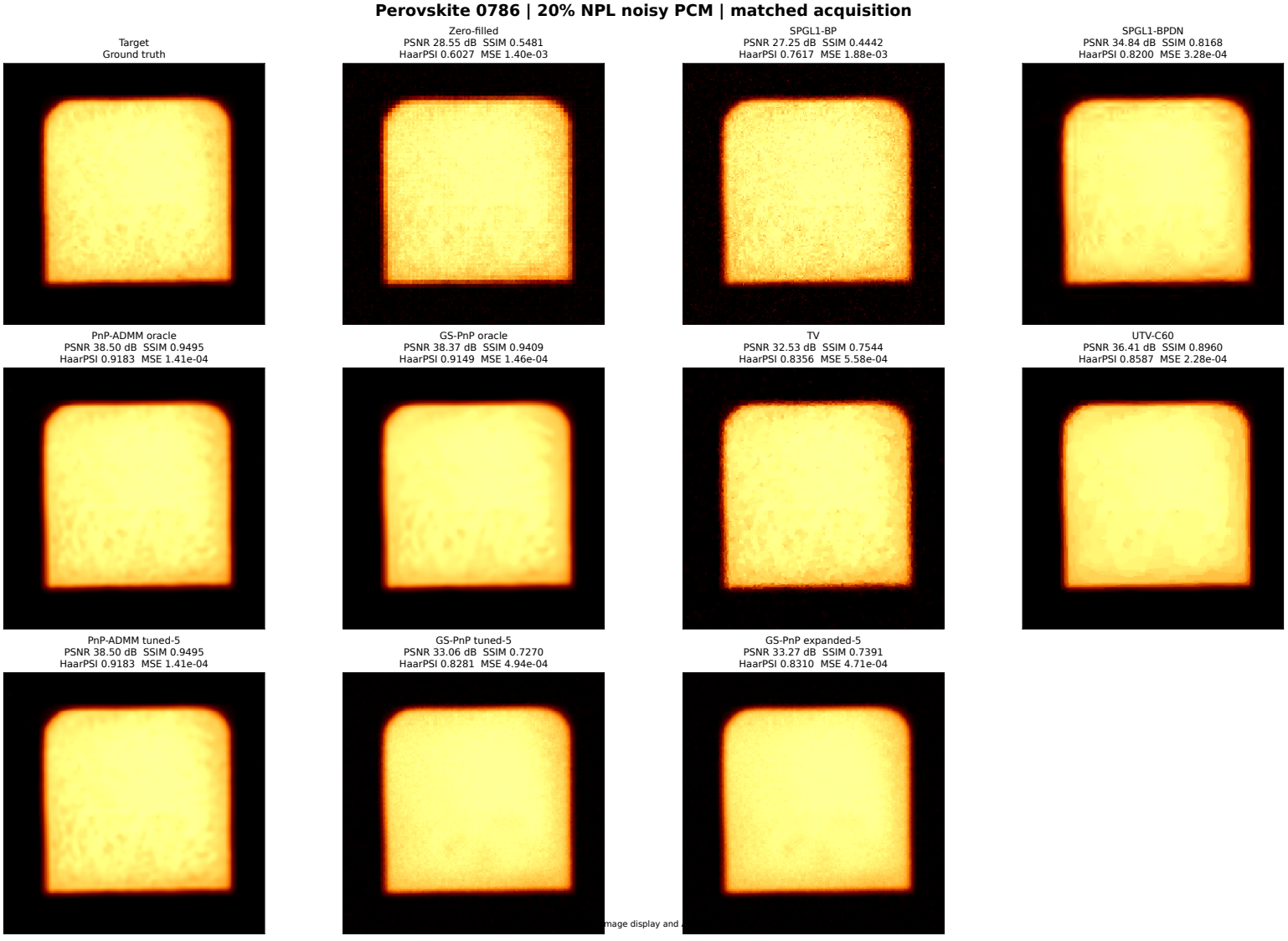
Table 7: Category mean PSNR in dB; 90 matched test cases per category. The full ten-method category metrics and their sample standard deviations are in the companion JSON and figure files.

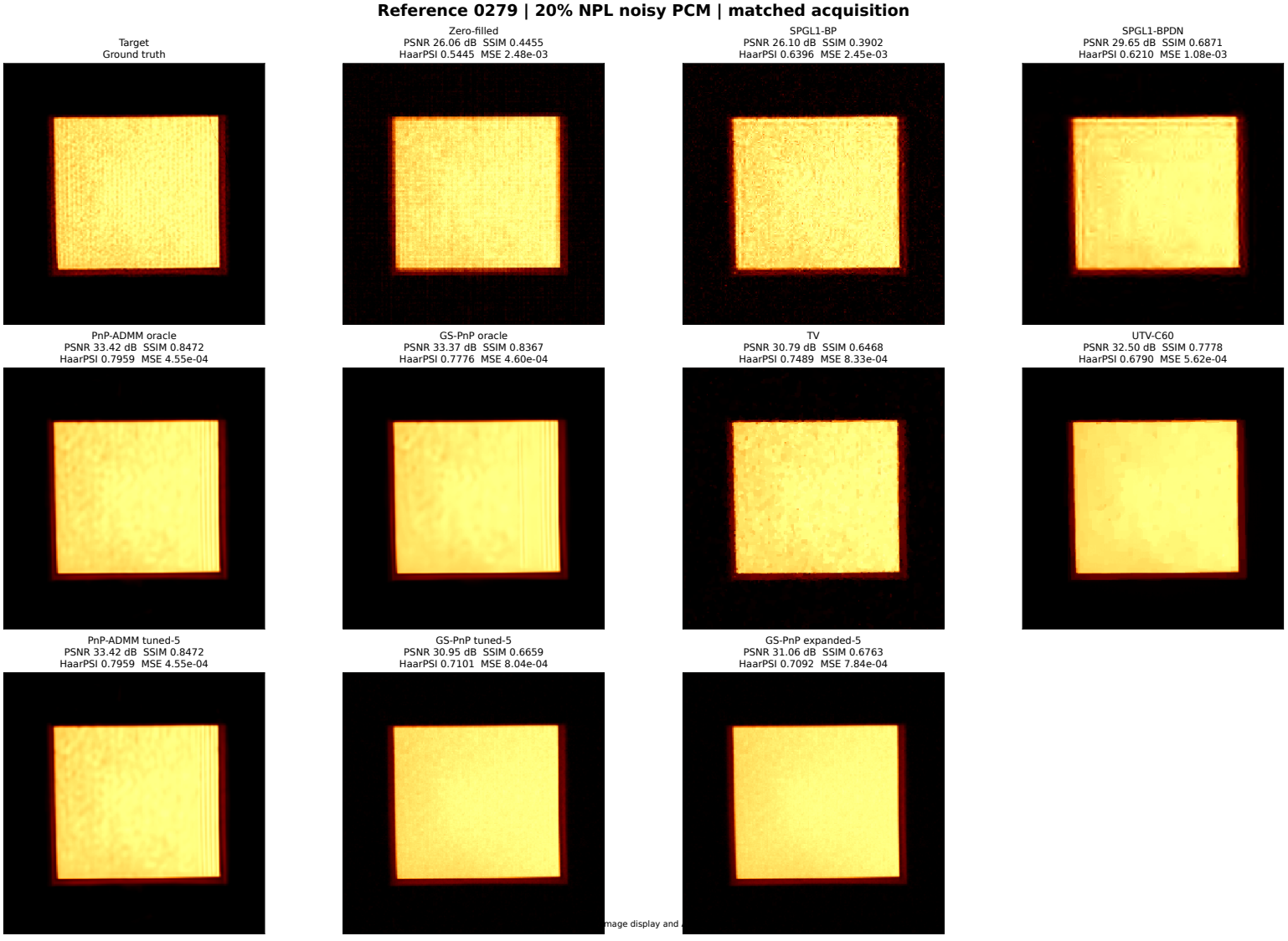
Five illustrative PCM reconstructions

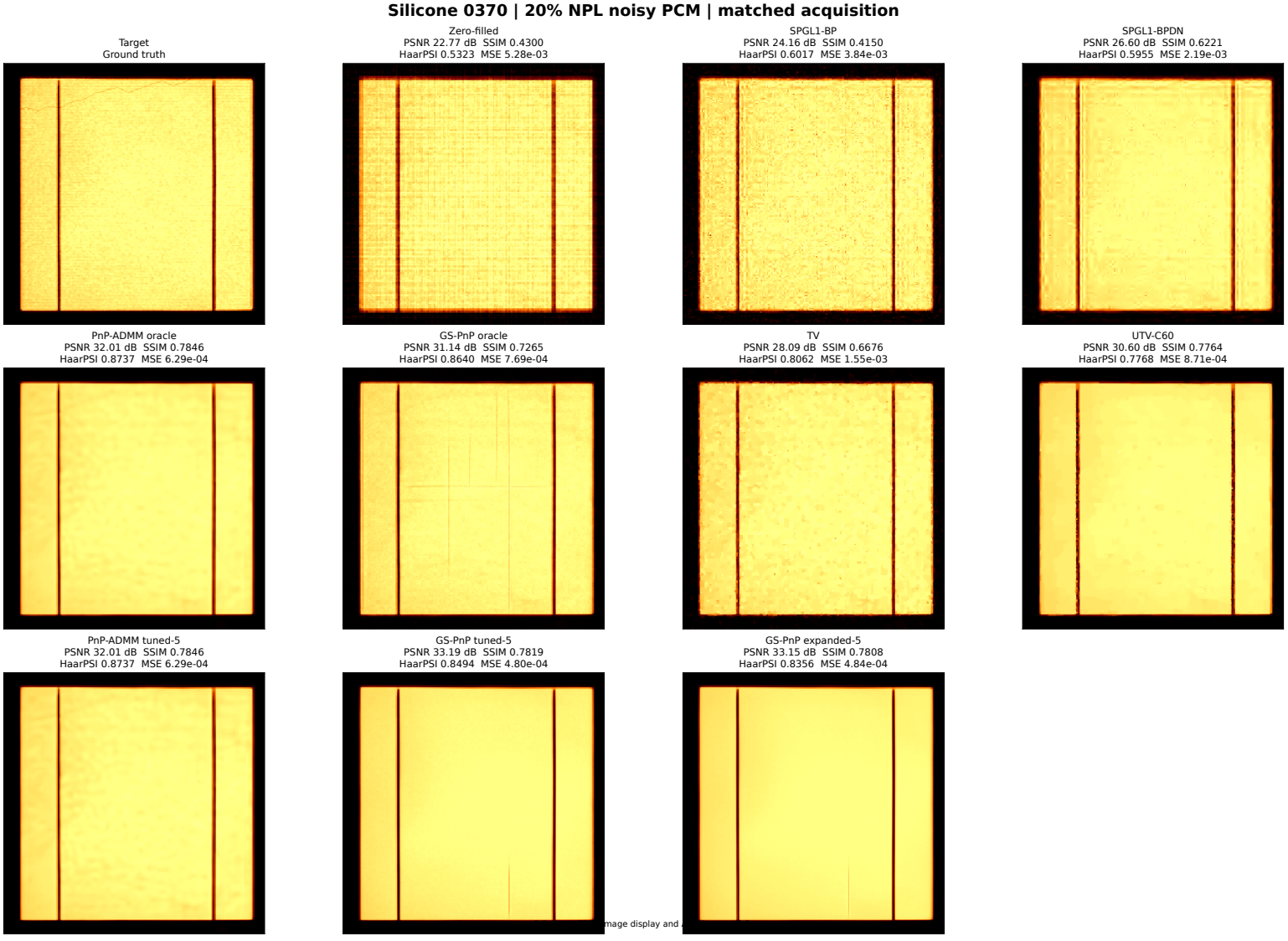
Five example images are shown using exactly the previously fixed case selection: CIGS 0279, Organic 0331, Perovskite 0786, Reference 0279 and Silicone 0370. Each comparison includes one target and all ten methods, displaying PSNR, SSIM, HaarPSI and MSE beside the image. Colour is AFM-hot capped at 0.8, with identical image intensity range $[0, 1]$ for every panel. These examples are illustrative and were not used to adjust hyperparameters.


CIGS 0279 | 20% NPL noisy PCM | matched acquisition
Target
Ground truth
Zero-filled
PSNR 26.51 dB SSIM 0.5039
HaarPSI 0.6453 MSE 2.23e-03
SPGL1-BP
PSNR 27.01 dB SSIM 0.4674
HaarPSI 0.7333 MSE 1.99e-03
SPGL1-BPDN
PSNR 32.31 dB SSIM 0.8341
HaarPSI 0.7670 MSE 5.88e-04
PnP-ADMM oracle
PSNR 40.27 dB SSIM 0.9877
HaarPSI 0.9597 MSE 9.39e-05
GS-PnP oracle
PSNR 37.63 dB SSIM 0.9838
HaarPSI 0.9586 MSE 1.72e-04
TV
PSNR 31.78 dB SSIM 0.7676
HaarPSI 0.8369 MSE 6.64e-04
UTV-C60
PSNR 33.42 dB SSIM 0.9372
HaarPSI 0.8287 MSE 4.55e-04
PnP-ADMM tuned-5
PSNR 40.27 dB SSIM 0.9877
HaarPSI 0.9597 MSE 9.39e-05
GS-PnP tuned-5
PSNR 35.70 dB SSIM 0.8774
HaarPSI 0.9121 MSE 2.69e-04
GS-PnP expanded-5
PSNR 35.92 dB SSIM 0.8864
HaarPSI 0.9144 MSE 2.56e-04


Organic 0331 | 20% NPL noisy PCM | matched acquisition
Target
Ground truth
Zero-filled
PSNR 24.46 dB SSIM 0.4570
HaarPSI 0.5363 MSE 3.58e-03
SPGL1-BP
PSNR 26.42 dB SSIM 0.4245
HaarPSI 0.6878 MSE 2.28e-03
SPGL1-BPDN
PSNR 32.31 dB SSIM 0.8019
HaarPSI 0.7152 MSE 5.88e-04
PnP-ADMM oracle
PSNR 38.82 dB SSIM 0.9464
HaarPSI 0.9226 MSE 1.31e-04
GS-PnP oracle
PSNR 39.39 dB SSIM 0.9512
HaarPSI 0.9252 MSE 1.15e-04
TV
PSNR 31.34 dB SSIM 0.7699
HaarPSI 0.8253 MSE 7.35e-04
UTV-C60
PSNR 36.33 dB SSIM 0.9385
HaarPSI 0.8826 MSE 2.33e-04
PnP-ADMM tuned-5
PSNR 38.82 dB SSIM 0.9464
HaarPSI 0.9226 MSE 1.31e-04
GS-PnP tuned-5
PSNR 34.82 dB SSIM 0.8188
HaarPSI 0.8935 MSE 3.29e-04
GS-PnP expanded-5
PSNR 35.05 dB SSIM 0.8289
HaarPSI 0.8955 MSE 3.12e-04


Perovskite 0786 | 20% NPL noisy PCM | matched acquisition
Target
Ground truth
Zero-filled
PSNR 28.55 dB SSIM 0.5481
HaarPSI 0.6027 MSE 1.40e-03
SPGL1-BP
PSNR 27.25 dB SSIM 0.4442
HaarPSI 0.7617 MSE 1.88e-03
SPGL1-BPDN
PSNR 34.84 dB SSIM 0.8168
HaarPSI 0.8200 MSE 3.28e-04
PnP-ADMM oracle
PSNR 38.50 dB SSIM 0.9495
HaarPSI 0.9183 MSE 1.41e-04
GS-PnP oracle
PSNR 38.37 dB SSIM 0.9409
HaarPSI 0.9149 MSE 1.46e-04
TV
PSNR 32.53 dB SSIM 0.7544
HaarPSI 0.8356 MSE 5.58e-04
UTV-C60
PSNR 36.41 dB SSIM 0.8960
HaarPSI 0.8587 MSE 2.28e-04
PnP-ADMM tuned-5
PSNR 38.50 dB SSIM 0.9495
HaarPSI 0.9183 MSE 1.41e-04
GS-PnP tuned-5
PSNR 33.06 dB SSIM 0.7270
HaarPSI 0.8281 MSE 4.94e-04
GS-PnP expanded-5
PSNR 33.27 dB SSIM 0.7391
HaarPSI 0.8310 MSE 4.71e-04


Reference 0279 | 20% NPL noisy PCM | matched acquisition
Target
Ground truth
Zero-filled
PSNR 26.06 dB SSIM 0.4455
HaarPSI 0.5445 MSE 2.48e-03
SPGL1-BP
PSNR 26.10 dB SSIM 0.3902
HaarPSI 0.6396 MSE 2.45e-03
SPGL1-BPDN
PSNR 29.65 dB SSIM 0.6871
HaarPSI 0.6210 MSE 1.08e-03
PnP-ADMM oracle
PSNR 33.42 dB SSIM 0.8472
HaarPSI 0.7959 MSE 4.55e-04
GS-PnP oracle
PSNR 33.37 dB SSIM 0.8367
HaarPSI 0.7776 MSE 4.60e-04
TV
PSNR 30.79 dB SSIM 0.6468
HaarPSI 0.7489 MSE 8.33e-04
UTV-C60
PSNR 32.50 dB SSIM 0.7778
HaarPSI 0.6790 MSE 5.62e-04
PnP-ADMM tuned-5
PSNR 33.42 dB SSIM 0.8472
HaarPSI 0.7959 MSE 4.55e-04
GS-PnP tuned-5
PSNR 30.95 dB SSIM 0.6659
HaarPSI 0.7101 MSE 8.04e-04
GS-PnP expanded-5
PSNR 31.06 dB SSIM 0.6763
HaarPSI 0.7092 MSE 7.84e-04


Silicone 0370 | 20% NPL noisy PCM | matched acquisition
Target
Ground truth
Zero-filled
PSNR 22.77 dB SSIM 0.4300
HaarPSI 0.5323 MSE 5.28e-03
SPGL1-BP
PSNR 24.16 dB SSIM 0.4150
HaarPSI 0.6017 MSE 3.84e-03
SPGL1-BPDN
PSNR 26.60 dB SSIM 0.6221
HaarPSI 0.5955 MSE 2.19e-03
PnP-ADMM oracle
PSNR 32.01 dB SSIM 0.7846
HaarPSI 0.8737 MSE 6.29e-04
GS-PnP oracle
PSNR 31.14 dB SSIM 0.7265
HaarPSI 0.8640 MSE 7.69e-04
TV
PSNR 28.09 dB SSIM 0.6676
HaarPSI 0.8062 MSE 1.55e-03
UTV-C60
PSNR 30.60 dB SSIM 0.7764
HaarPSI 0.7768 MSE 8.71e-04
PnP-ADMM tuned-5
PSNR 32.01 dB SSIM 0.7846
HaarPSI 0.8737 MSE 6.29e-04
GS-PnP tuned-5
PSNR 33.19 dB SSIM 0.7819
HaarPSI 0.8494 MSE 4.80e-04
GS-PnP expanded-5
PSNR 33.15 dB SSIM 0.7808
HaarPSI 0.8356 MSE 4.84e-04

6 Conclusion

The expanded five-image validation SSDU sweep is complete, and the winning configuration was frozen before the additional 450-image reconstruction evaluation. The measurement-only SSDU loss fell from 0.004365434 to 0.004361004 (0.10 percent reduction). The validated candidate is `gs_exp_002`, with $\lambda = 0.100$, $\sigma = 0.075$ and initial $\tau_0 = 50.0000$. The selected candidate lies at new upper search bounds as follows: lambda no, sigma no, initial step no.

The expanded configuration gives mean PSNR 32.563 dB, compared with 32.590 dB for the original SSDU-5 GS-PnP and 35.110 dB for the supervised-validation oracle. The paired expanded-minus-original PSNR difference is -0.027 dB (sample SD 0.531 dB), while the difference against the oracle is -2.547 dB. All test parameters were frozen before scoring these images.

These observations show the behaviour of this finite, preregistered SSDU grid. They do not prove a global optimum. The old test set had already been inspected in the broader research project, so this is a retrospective controlled test. Its results must not be used to make further unrecorded hyperparameter changes. PCM noise correlation and the undersampled Hadamard nullspace remain important limitations.

Primary papers. Yaman et al., *Self-Supervised Learning of Physics-Guided Reconstruction Neural Networks without Fully-Sampled Reference Data*, Magnetic Resonance in Medicine (2020), acquired measurement partitioning: `https://doi.org/10.1002/mrm.28378`.

Hurault, Leclaire and Papadakis, *Gradient Step Denoiser for Convergent Plug-and-Play*, ICLR (2022), gradient-step update and backtracking: `https://openreview.net/pdf?id=fPhKeld3Okz`.